

Methods of fixing DRCs :-

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1. manual editing of netlist :

↳ created a copy of the netlist and edited it to fix the DRC.



We performed scan insertion for this edited netlist.

2. Using tessel commands :

(create-instance, delete-connection, create-connections)

In method 1, we had added the muxes by manually editing the netlist and we have fixed the DRC.

In method 2, we will use the above 3 commands to add the mux into our netlist and fix the DRC.

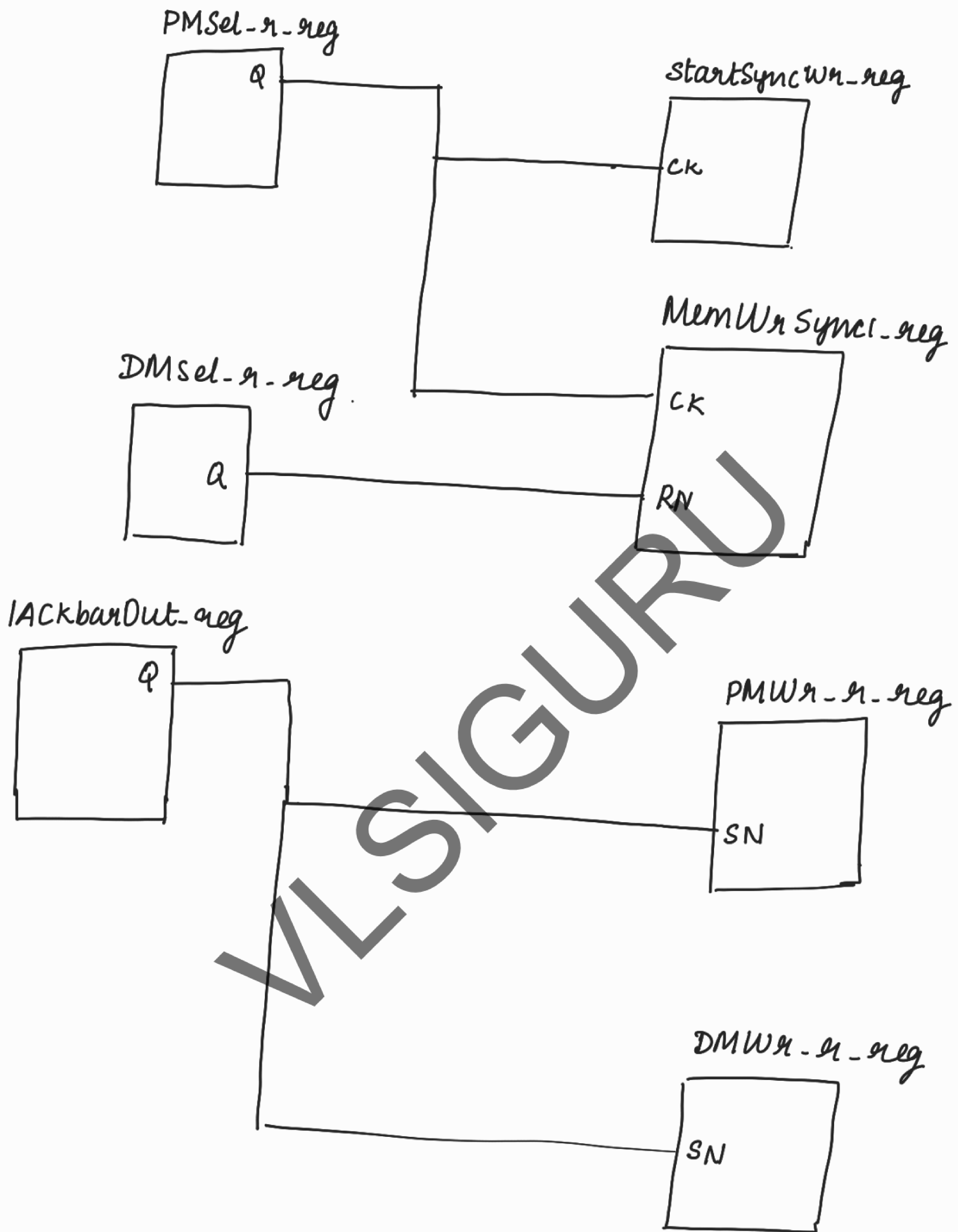


updated netlist (netlist after adding the muxes)

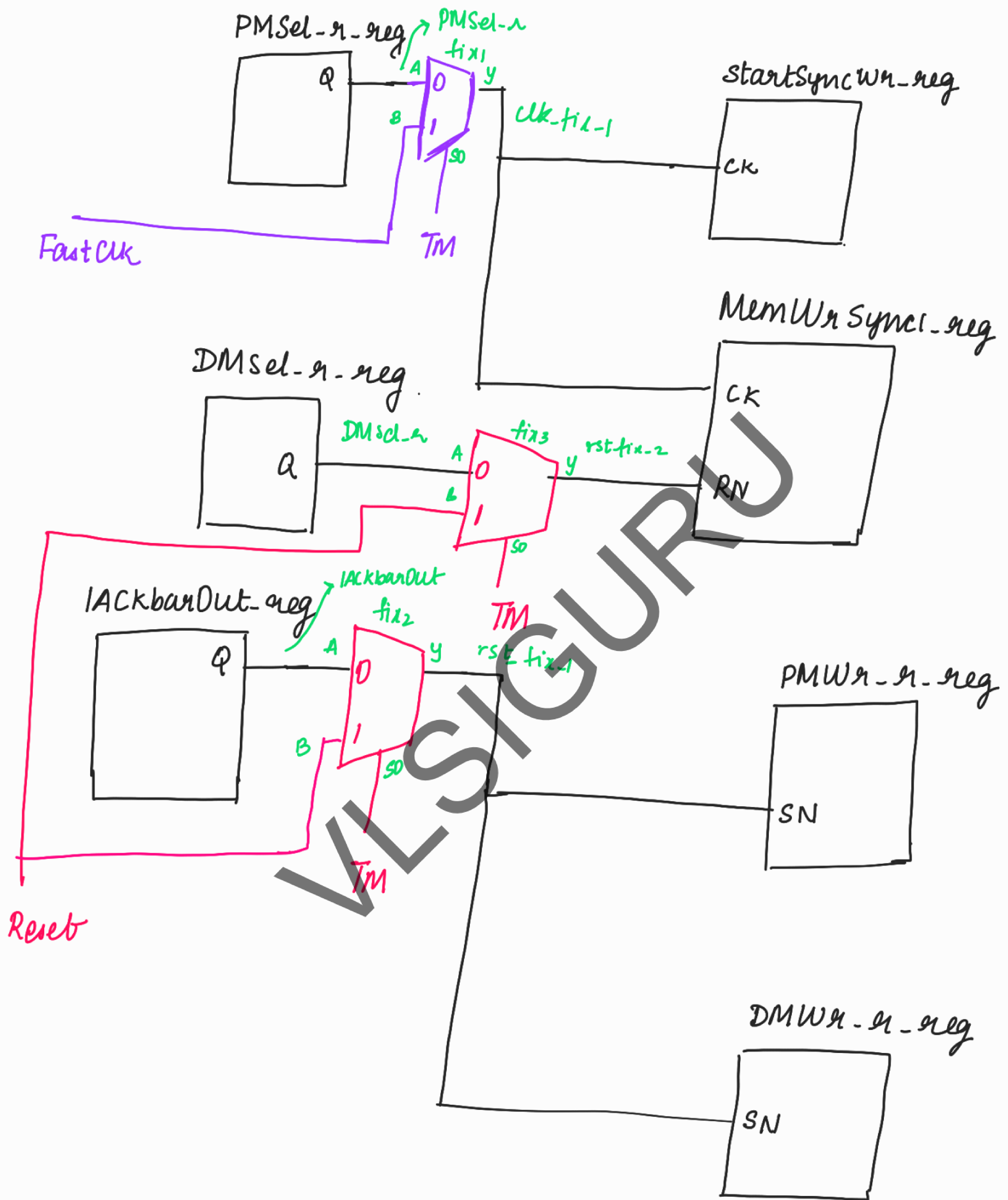


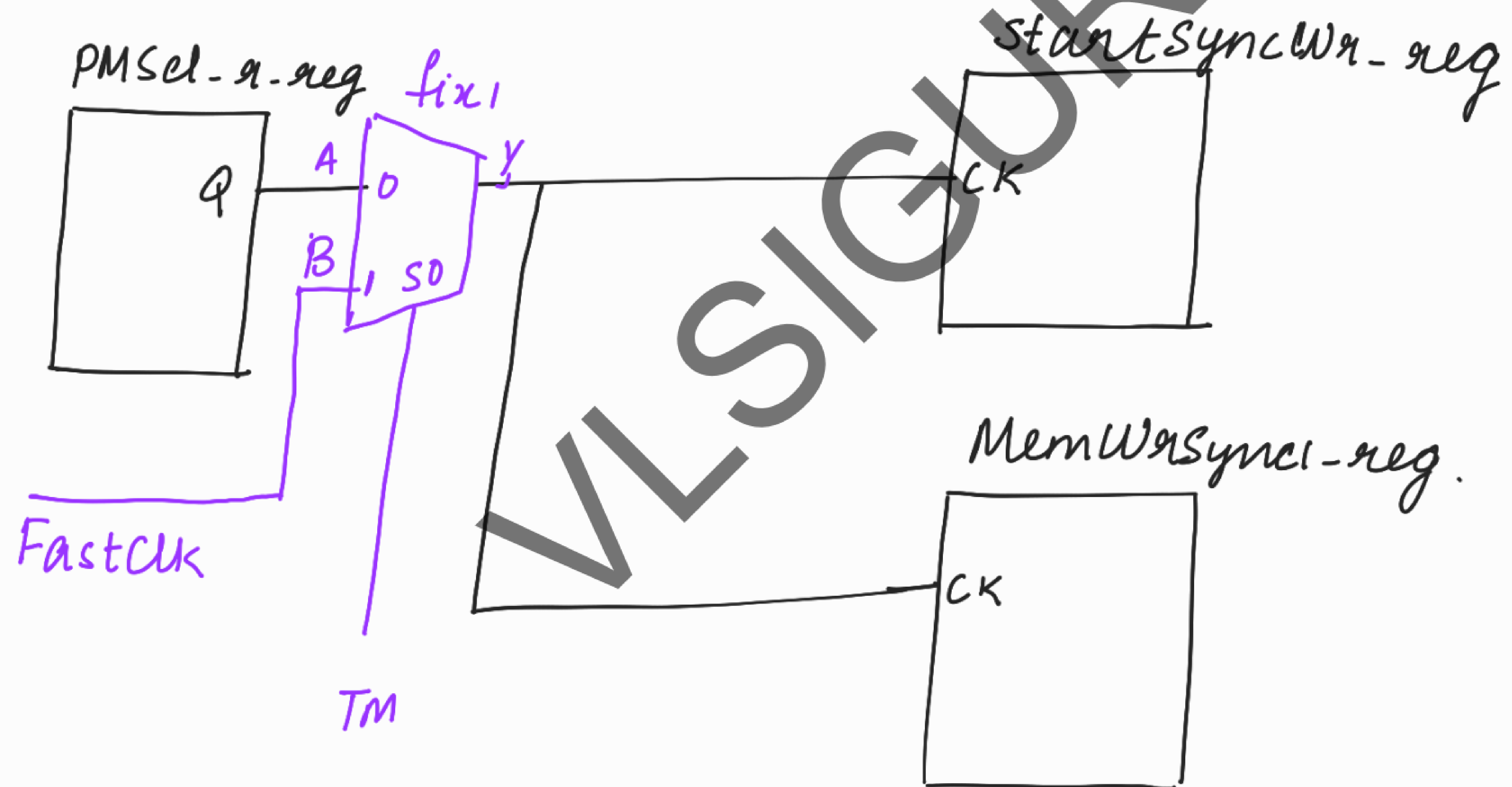
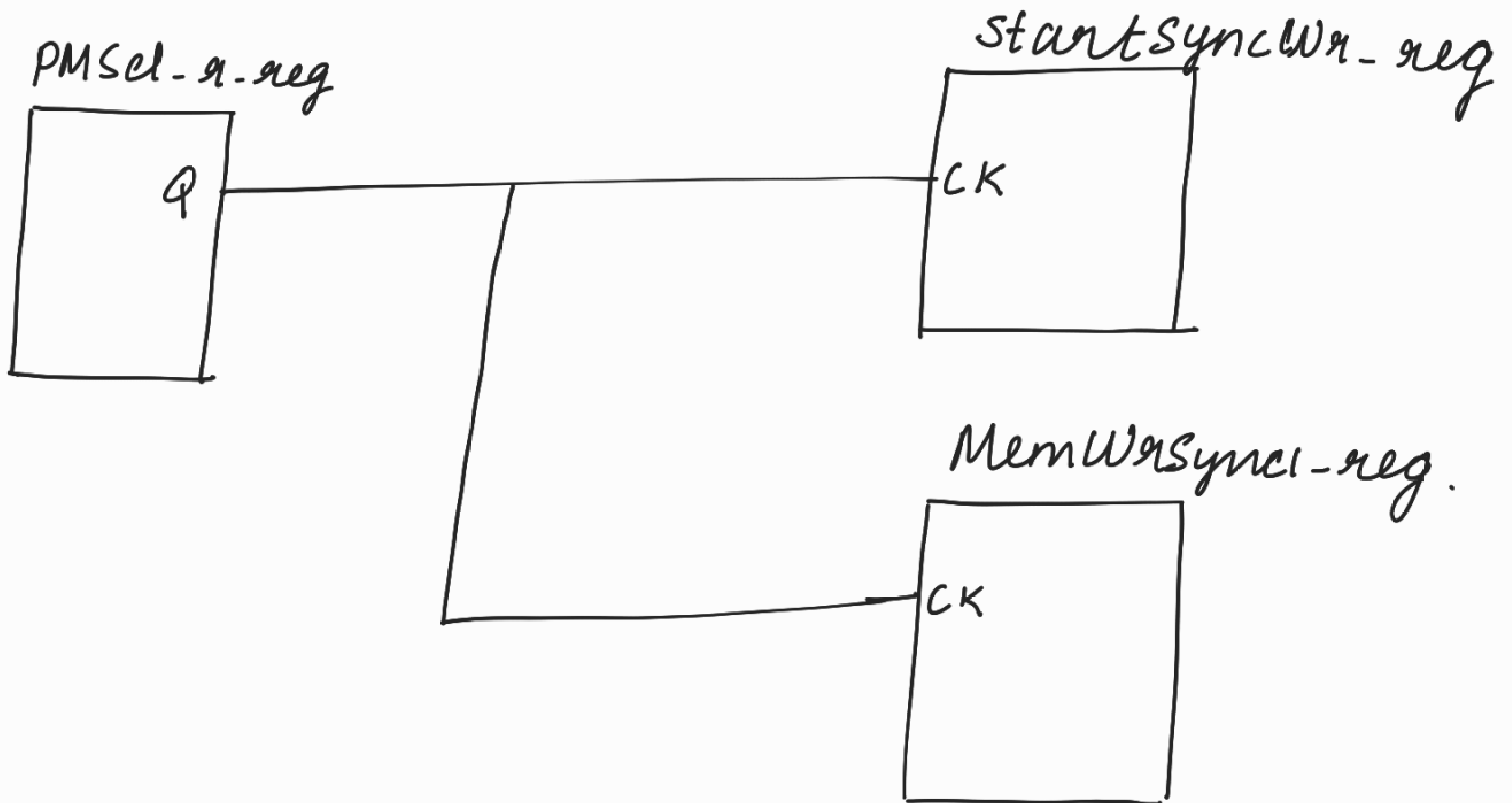
Perform scan insertion for this updated netlist.

COMPLETE FIGURE :-



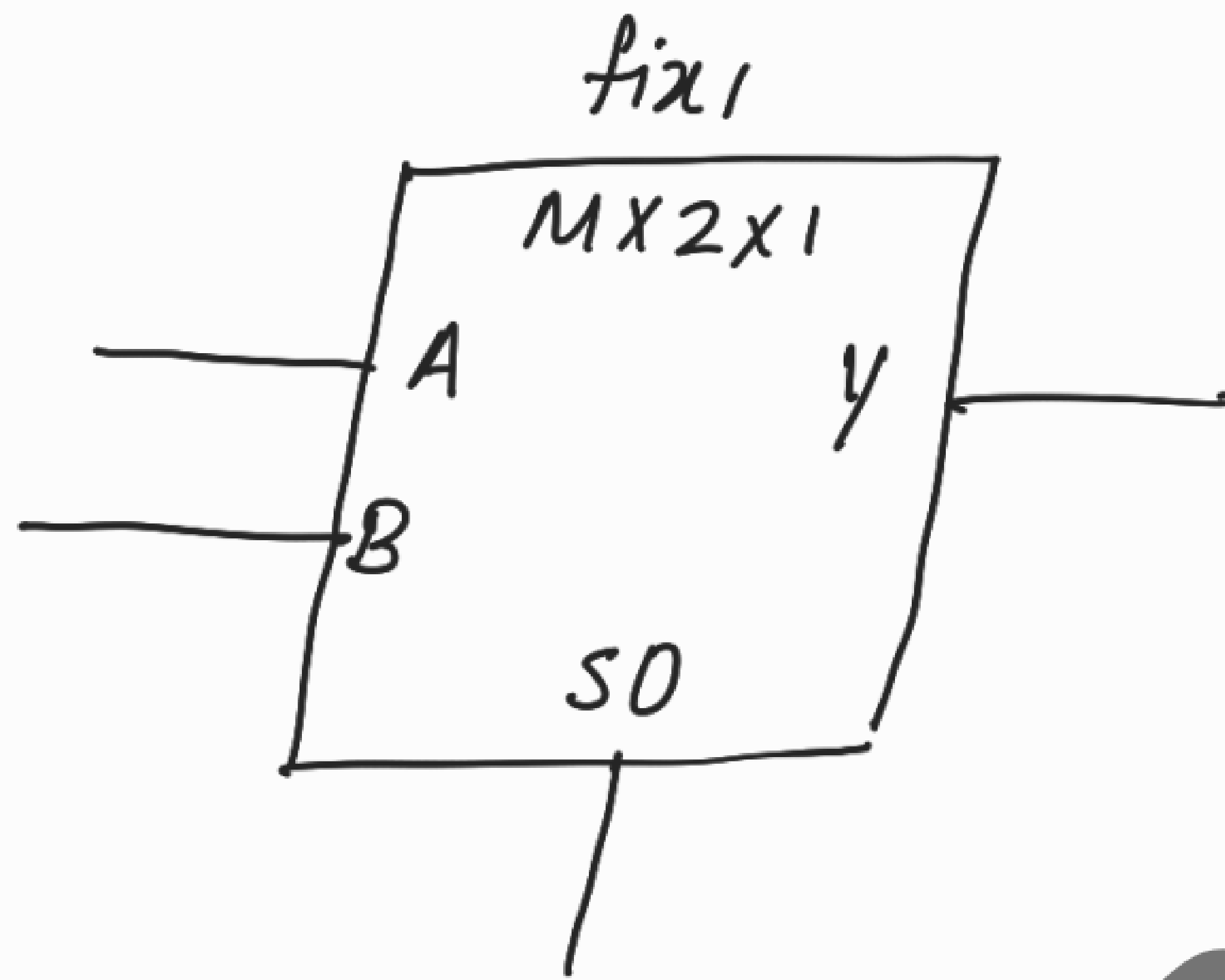
Fix:





(i) create-instance <name> -of-module <lib model name>

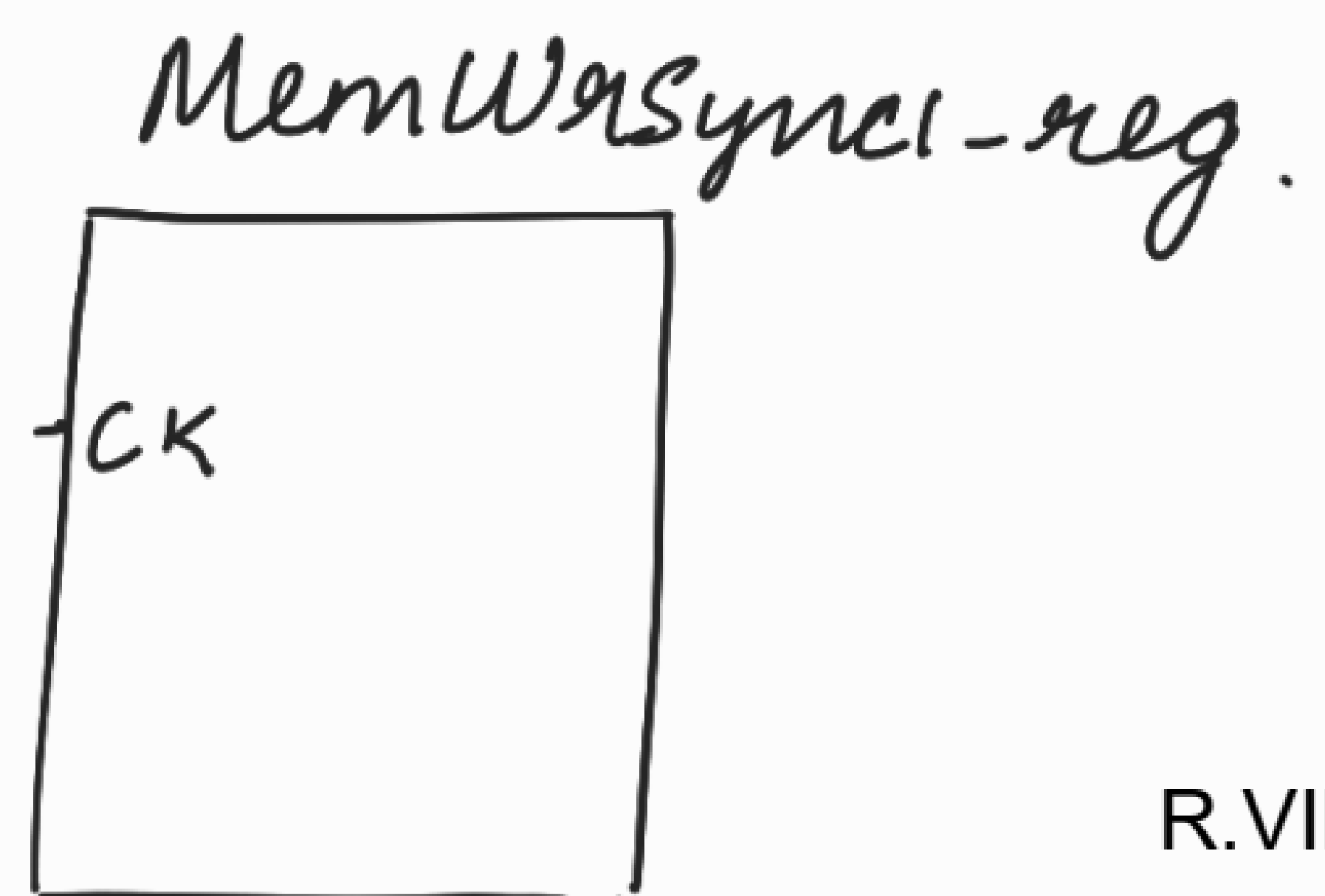
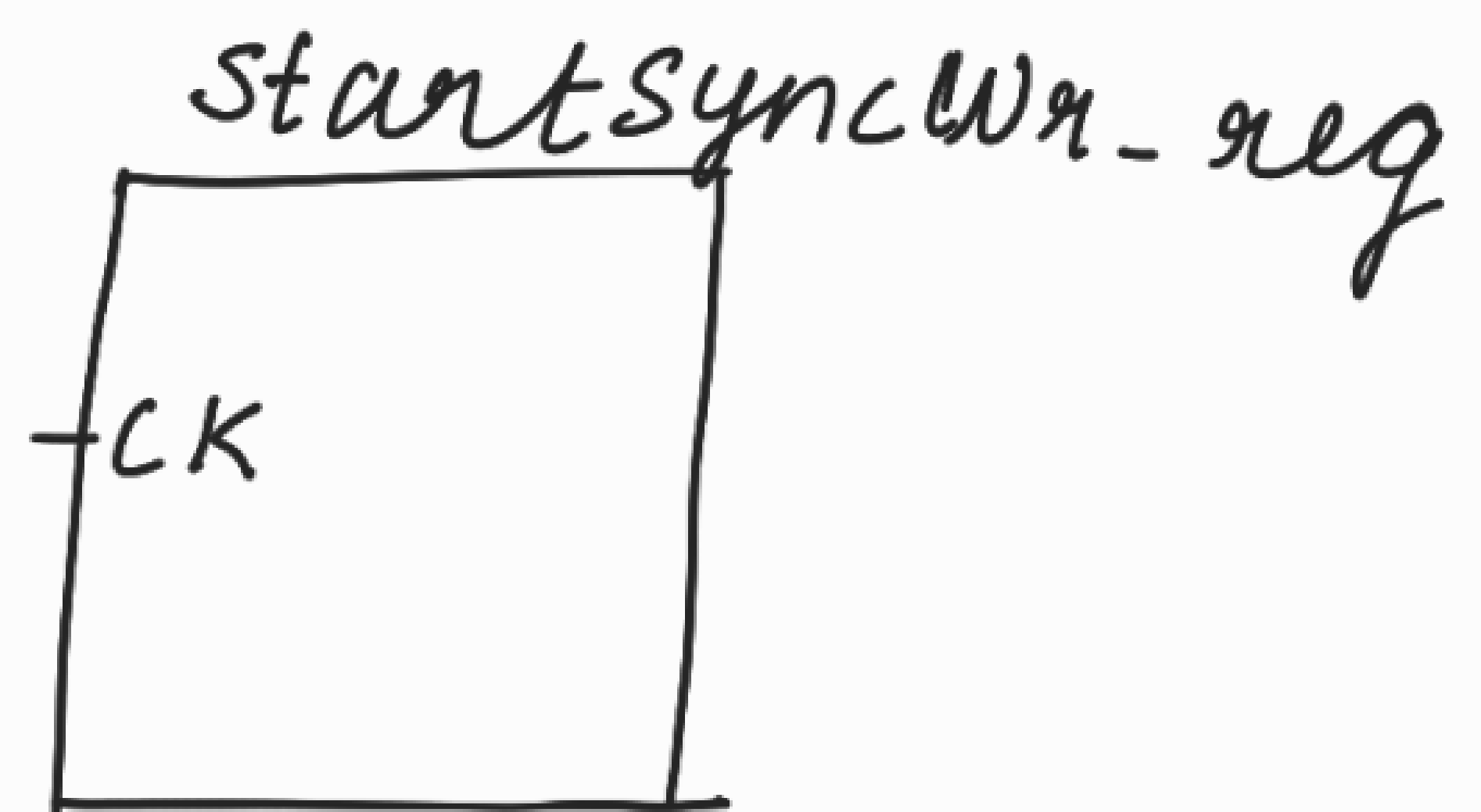
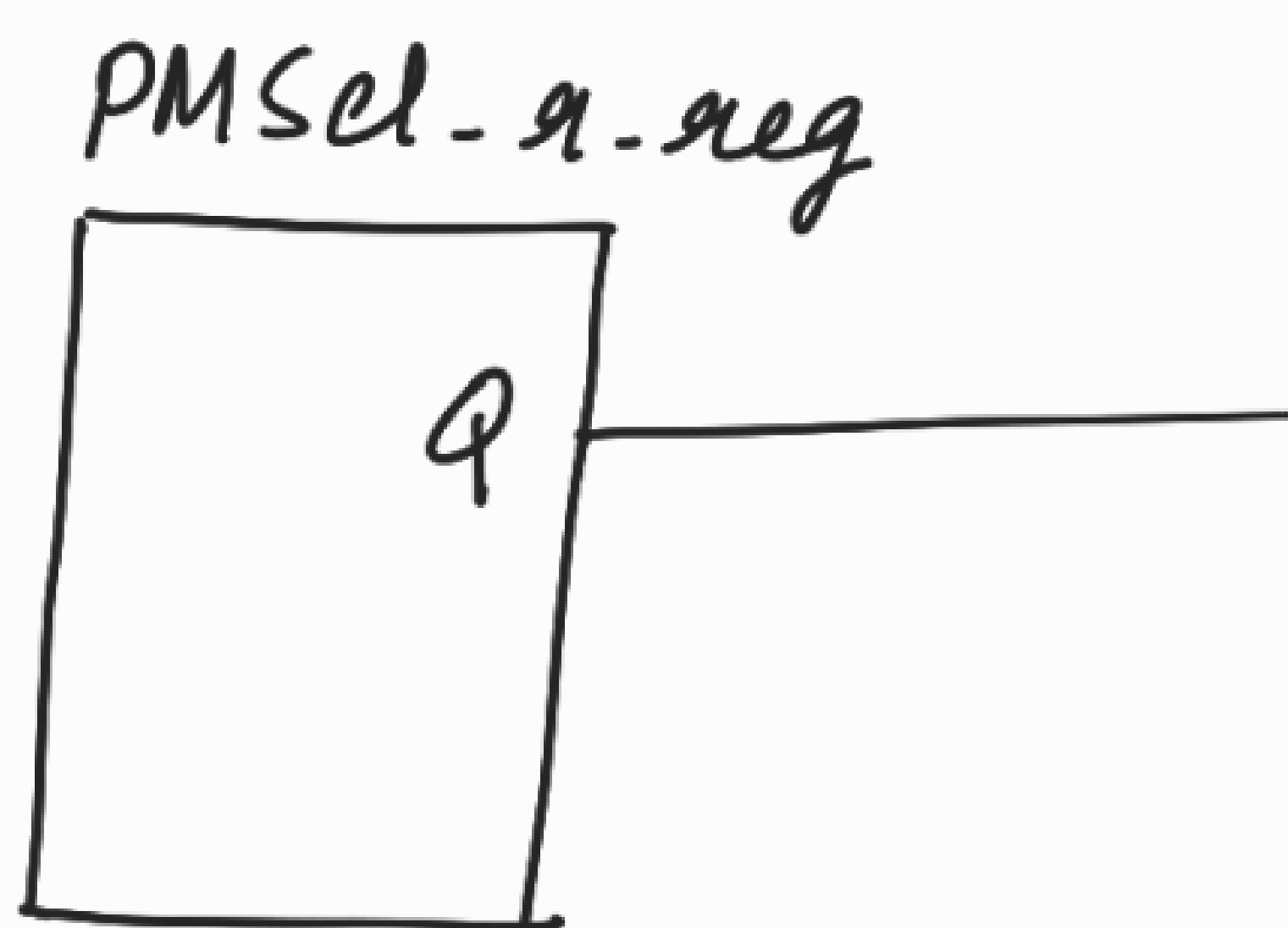
create-instance fix1 -of-module MX2X1

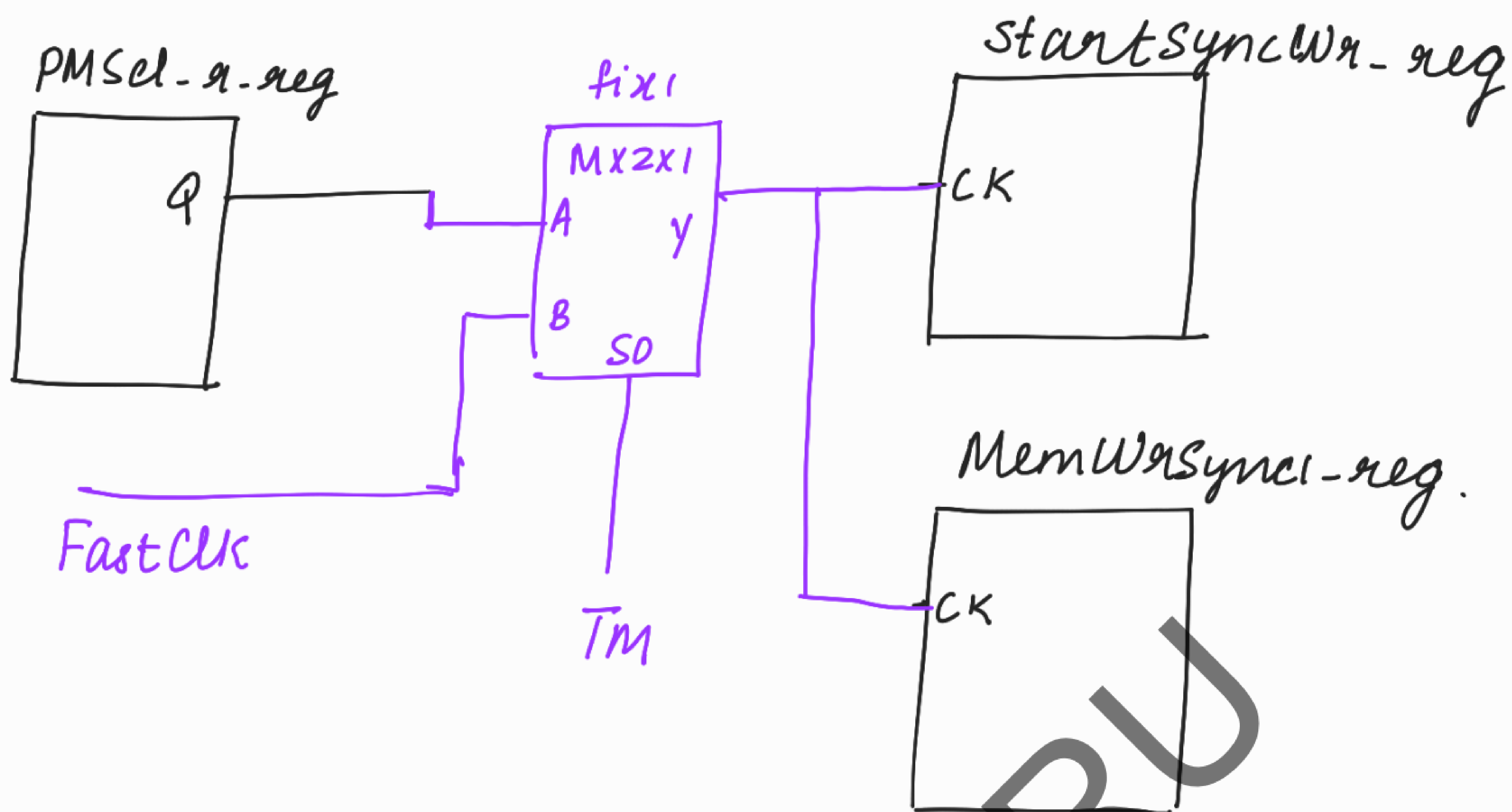


(ii) delete-connection <which connection>

delete-connection /startSyncWr-reg/CK

delete-connection /MemWrSyncr-reg/CK





(iii) create-connection <from> <to>

create-connection	fix1/A	/PMSel-a-reg/Q
create-connection	fix1/B	FastClk
create-connection	fix1/SD	TM ← See * This is a top level port. We need to create it 1st.
create-connection	fix1/y	/StartSyncWn-reg/CK
create-connection	fix1/y	/MemWnSync1-reg/CK.

* create-port <name> -direction <input/output>

create-port TM -direction input

NOTE:- All the above commands is written into a file (fix.tcl)

This method involves 2 runs

1st run :- (i) Source the fix.tcl (insertion mode)

(ii) Save the changes to a file

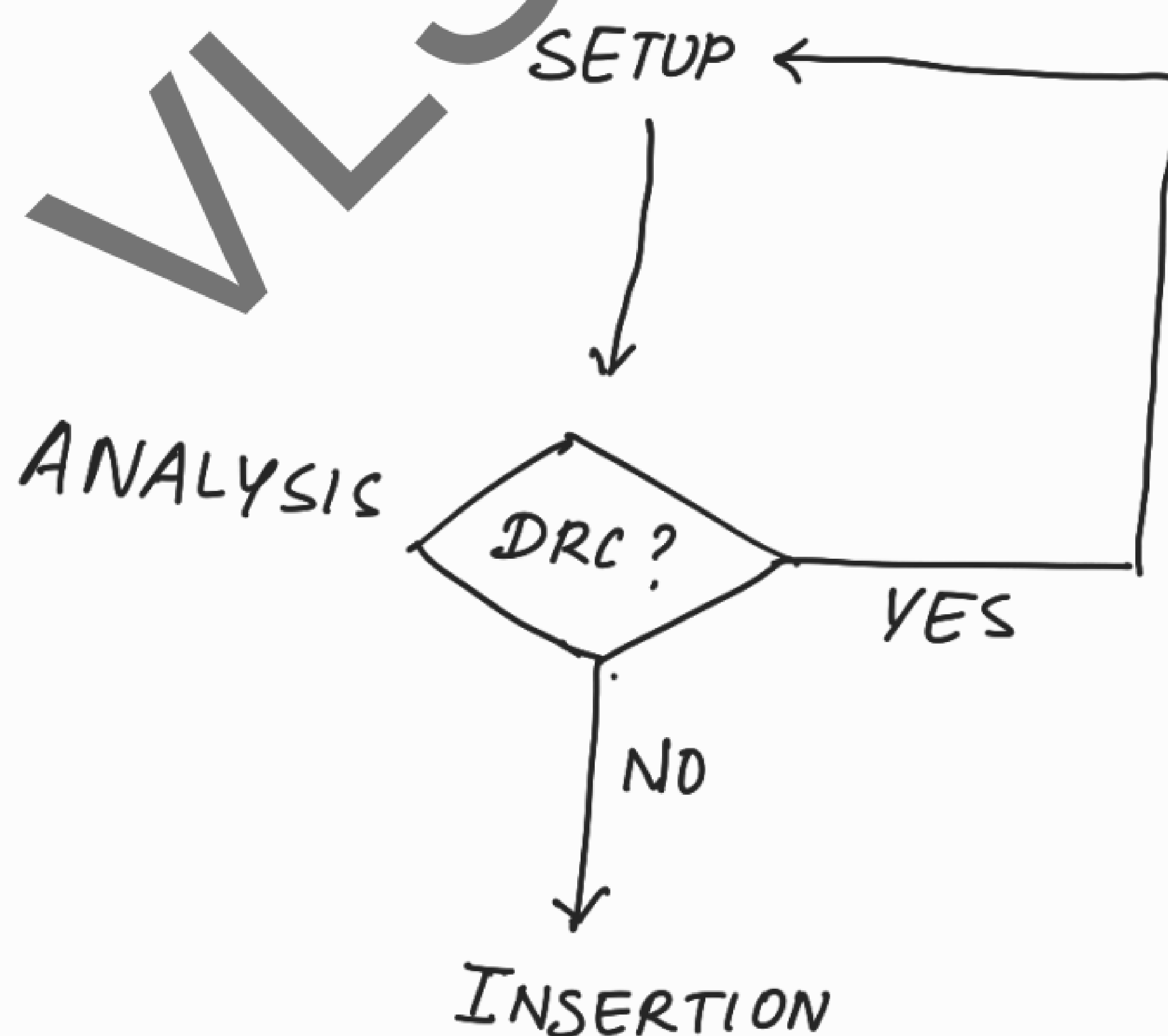
(case1-fix.v)

↑ original + merges added to
netlist fix the DRC.

2nd run :- Perform scan insertion for case1-fix.v file.

3. autofix command :-

↳ the tool will automatically add the logic and fix the DRC.



Setup mode :- set-test-logic -clock on -set on
-reset on

1st preference :- autofix method

(set-test-logic -clock on -set on -reset on)

→ tool will automatically fix the DRC without human effort.

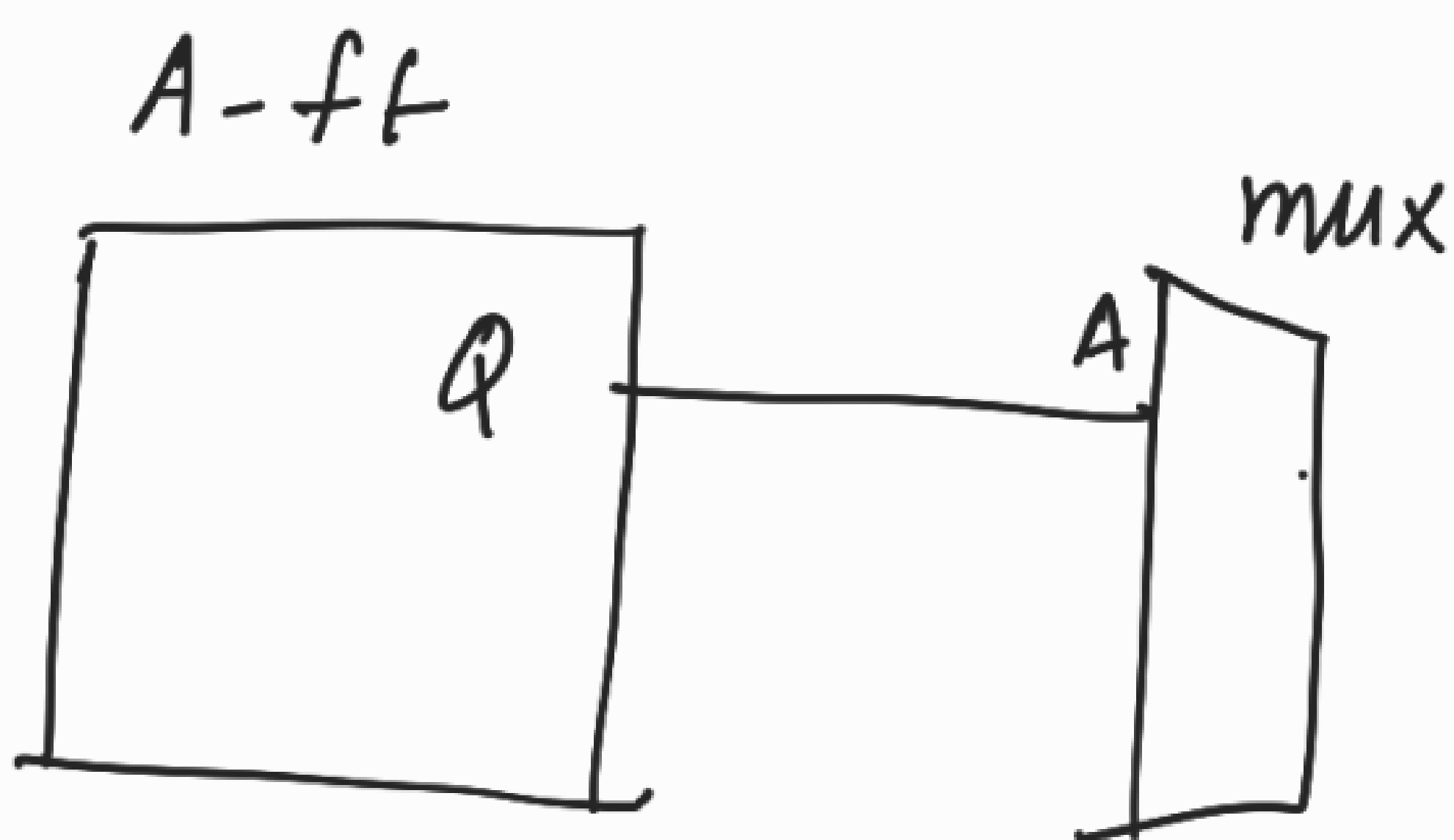
→ errors can be avoided.

2nd preference :- create-instance, delete-connection, create-connection

Note :- Why

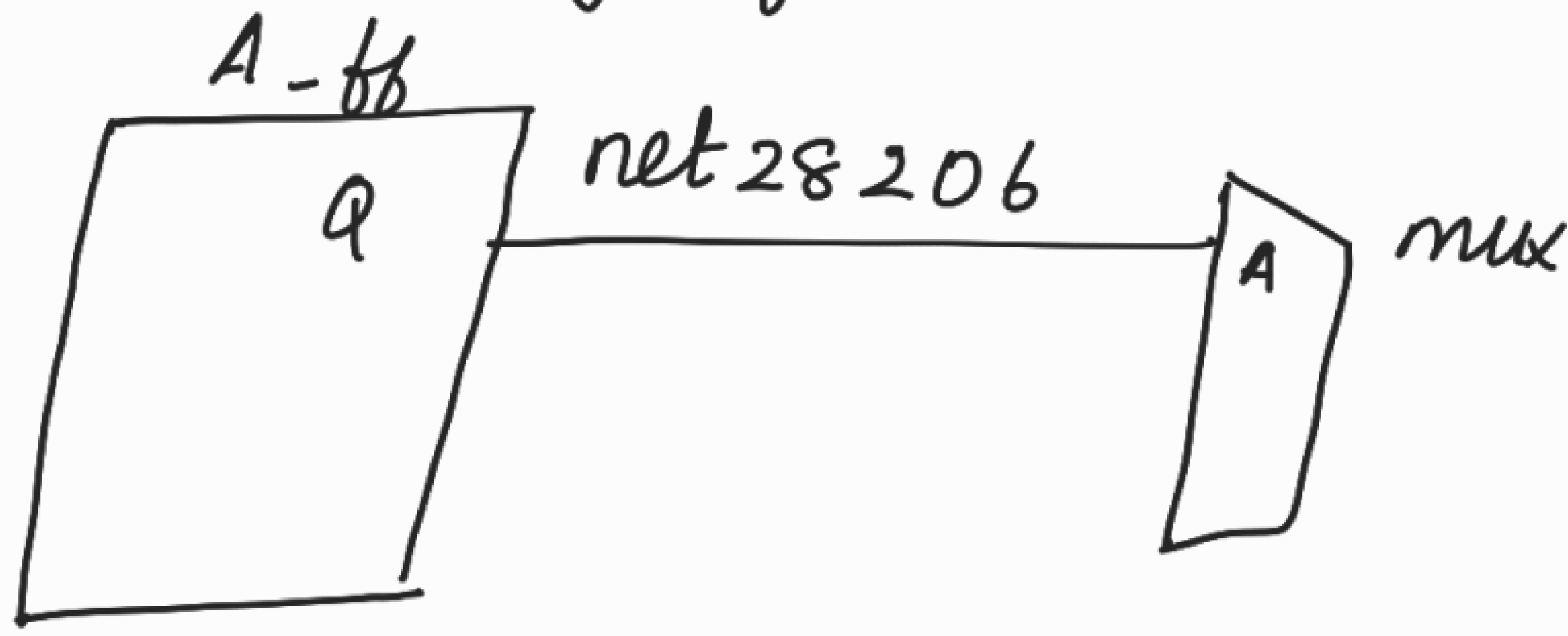
↓
create-instance, delete-connection,
create-connection

is preferred over manual hacking netlist?



create-connection A-ff/Q mux/A

manual editing of netlist



MX2x1 mux (. A (net28206))

more error prone

i.e. instead of net28206 if you give net2806, then wrong connection will be formed.

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